This is a manual for installing OpenCV with CUDA support on Jetson Nano.

Prerequisite:

1. Properly flashed SD card with the latest jetpack
2. Ethernet or WiFi connection
3. Admin permission on the Jetson's Ubuntu GUI
4. Keyboard, mouse, and monitor connected to the Jetson

Step-1: Removing Old OpenCV instance

Jetpack 4.6.6 comes with OpenCV 3.2.0, but CUDA support is turned off. The only way to turn CUDA support on is to build OpenCV 4.5.0 from the source code. That's why any previous OpenCV instance must be removed. To remove OpenCV, run:

```
sudo apt purge -y libopencv* python3-opencv
```

then

```
sudo apt purge -y libopencv* python3-opencv
```

Step-2: Install and Build Dependencies

Run:

```
sudo apt update
```

then

```
sudo apt install -y build-essential cmake git unzip pkg-config \
   libgtk-3-dev libjpeg-dev libpng-dev libtiff-dev \
   libavcodec-dev libavformat-dev libswscale-dev libv4l-dev \
   libxvidcore-dev libx264-dev libtbb2 libtbb-dev libdc1394-22-dev \
   libgstreamer1.0-dev libgstreamer-plugins-base1.0-dev \
   python3-dev python3-numpy
```

Step-3: Add Swap Space

This manual assumes the Jetson already has a swap space created during flashing the SD card as instructed in Week-12 report.

Step-4: Clone OpenCV and Contrib

```
cd ~
git clone https://github.com/opencv/opencv.git
cd opencv
git checkout 4.5.0
cd ..
git clone https://github.com/opencv/opencv_contrib.git
cd opencv_contrib
git checkout 4.5.0
```

Step-5: Configure with CMake

Run:

cd ~/opencv

then

mkdir build && cd build

then

```
cmake -D CMAKE_BUILD_TYPE=RELEASE \
    -D CMAKE_INSTALL_PREFIX=/usr/local \
    -D OPENCV_EXTRA_MODULES_PATH=~/opencv_contrib/modules \
    -D WITH_CUDA=ON \
    -D ENABLE_FAST_MATH=ON \
    -D CUDA_FAST_MATH=ON \
    -D WITH_CUBLAS=ON \
    -D WITH_GSTREAMER=ON \
    -D WITH_V4L=ON \
    -D BUILD_opencv_python3=ON \
    -D BUILD_EXAMPLES=OFF ..
```

Step-6: Building OpenCV(takes around 4hours)

Run:

make -j4

then

sudo make install

then

sudo ldconfig

Step-7: Verification

Run:

python3

then

import cv2

then

print(cv2.__version__)

or

print(cv2.getBuildInformation())